

Spacecraft Re-entry



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 **Spacecraft Re-entry Supplementary Materials**

1. Introduction

Spacecraft re-entry fundamentally is a massive kinetic energy management problem. This is because the spacecraft moves at over 20 Mach (over 7,000 m/s), and as it enters the thick atmosphere, that kinetic energy gets converted into extreme heat, which is primarily due to the immense compression that the air experiences. Our project explores how heat shields violently convert that kinetic energy into heat and how heat shields protect the spacecraft. We do this by first learning the main thermal-fluid concepts involved and then by applying those ideas in a simplified model of an ablative heat shield.

2. Thermal-Fluid Physics We Needed to Learn

Before building our simplified heat-shield model, we first needed to understand why spacecraft re-entry is a thermal-fluid systems problem. When a spacecraft returns to Earth, it enters the atmosphere at hypersonic speed. At those speeds, the air in front of the vehicle cannot move out of the way smoothly. Instead, the flow compresses and forms a shock layer around the vehicle. This shock layer converts some of the spacecraft's kinetic energy into thermal energy in the gas near the surface, creating an intense heating environment. NASA's aerothermodynamics notes describe aerothermodynamics as the process of predicting the heating environment experienced by an entry vehicle, and they connect that prediction directly to thermal protection system design.

One of the most important ideas we learned was stagnation-point heating. The stagnation point is the location near the nose of the vehicle where the incoming flow is brought nearly to rest. Because the flow slows down strongly there, the stagnation region usually experiences the highest heating. This is why the shape of the spacecraft matters. A blunt body creates a detached shock wave in front of the vehicle, which spreads the heating over a larger region and reduces the peak heat flux compared with a sharp body. A common engineering relationship for stagnation-point convective heating shows that heat flux increases strongly with velocity and decreases as the nose radius increases:

$$q'' \propto \sqrt{\frac{\rho}{R_n}} V^3$$

This relationship helped us understand why re-entry heating is so sensitive to speed. Since velocity is cubed, even a small increase in re-entry velocity can cause a much larger increase in heat flux. It also shows why a larger nose radius can reduce heating by making the vehicle more blunt.

We also needed to separate the different modes of heat transfer. First, convection transfers heat from the hot shock-layer gas to the surface of the heat shield. Second, radiation matters because excited atoms and molecules in the shock layer can radiate energy toward the vehicle, while the hot heat-shield surface can also reradiate energy away. Third, conduction determines how much of the surface heat travels through the heat shield material toward the spacecraft structure. This conductive part is especially important for our model because our final engineering question is about the back-wall temperature: even if the outside surface gets extremely hot, the inside surface must stay below a safe limit.

The heat shield is not just a wall that “blocks heat.” It is an energy-management system. Reusable heat shields, such as ceramic tiles, mainly work by having low thermal conductivity and by reradiating heat from the surface. Ablative heat shields work differently. In an ablative material, some of the heat goes into changing, charring, decomposing, melting, sublimating, or removing material from the surface. NASA’s ablative TPS (Thermal Protection System) material notes describe ablation as thermal protection through material consumption, meaning the material sacrifices itself to carry energy away from the spacecraft.

A full model of an ablative heat shield would be very complicated. It would require modeling surface recession, pyrolysis gases, char formation, changing material properties, radiation, convection, and transient conduction all at the same time. For a two-week class project, that would be too much to model accurately. Instead, we used this learning section to understand the real physics qualitatively, then simplified the application into a one-dimensional transient heat conduction model. In our model, the heat shield is treated as a slab exposed to a simplified re-entry heat-flux profile. This lets us focus on a clear thermal-fluid question: finding the total ablation of the spacecraft during re-entry?

Overall, the physics we learned shows that re-entry heating is not caused by simple friction alone. It is caused by a coupled aerodynamics and heat-transfer problem involving hypersonic flow, shock-layer heating, radiation, surface energy balance, and conduction through the heat shield. This background gives us the foundation for the application part of the project, where we turn the physics into a simplified model for heat shield thickness.

3. Application: Simplified Re-entry and Heating Model

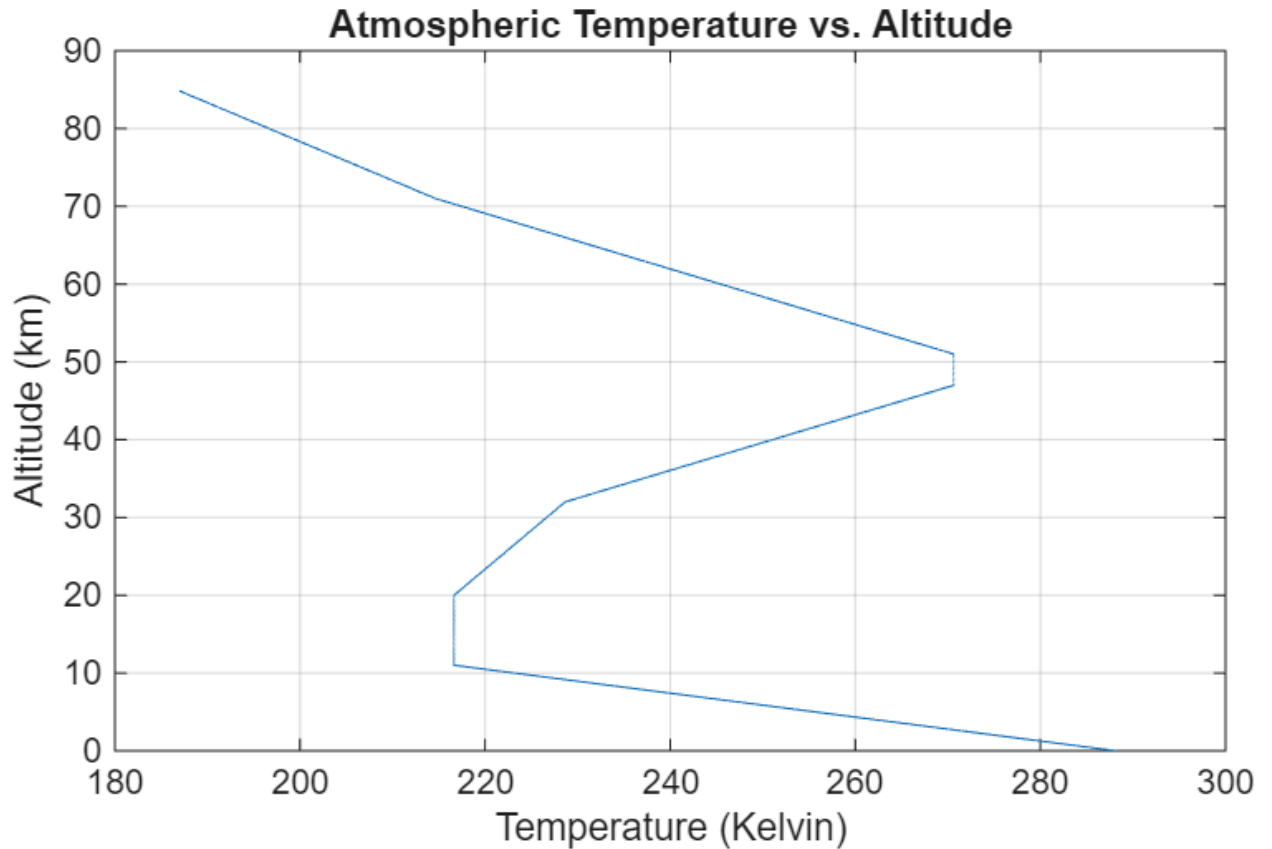
After learning the main thermal-fluid physics behind spacecraft re-entry, we applied those ideas in a simplified engineering model. The purpose of this model was to estimate how atmospheric conditions, vehicle trajectory, and aerodynamic heating change as a spacecraft descends through the atmosphere. Rather than trying to fully model every aspect of a real re-entry vehicle, we focused on a manageable first-order model that captures the main relationships between altitude, density, velocity, and stagnation-point heating.

Our approach had five main parts. First, we modeled how atmospheric properties change with altitude. Second, we used a simplified trajectory model to estimate how the spacecraft's velocity changes as it descends. Third, we used the Sutton-Graves stagnation-point heating relation to estimate the aerodynamic heating experienced by the spacecraft. And fourth, we used Lee's heat transfer distribution equation to relate the curvature of the nose to the heat using the stagnation point heating. Finally, fifth, using this heating profile to make a very simple approximation of the amount of ablation that occurs

We wanted to know how atmospheric density, spacecraft velocity, and altitude combine to determine the aerodynamic heating experienced during re-entry.

3.1 Atmospheric Model

The first part of the code constructs a layered model of the atmosphere from sea level to approximately 84.9 km altitude. Each atmospheric layer is assigned a lapse rate, base altitude, and base temperature. The following correlation between height and temperature was created, not accounting for different locations, times of day, or seasons.



Using this temperature profile, we calculated how pressure changes with altitude using the barometric formula

$$P(H) = P_0 \cdot \left(\frac{T_0}{T_0 + L \cdot (H - H_0)} \right)^{\left(\frac{g_0 \cdot M}{R \cdot L_0} \right)}$$

Equation for finding the pressure using a given temperature gradient

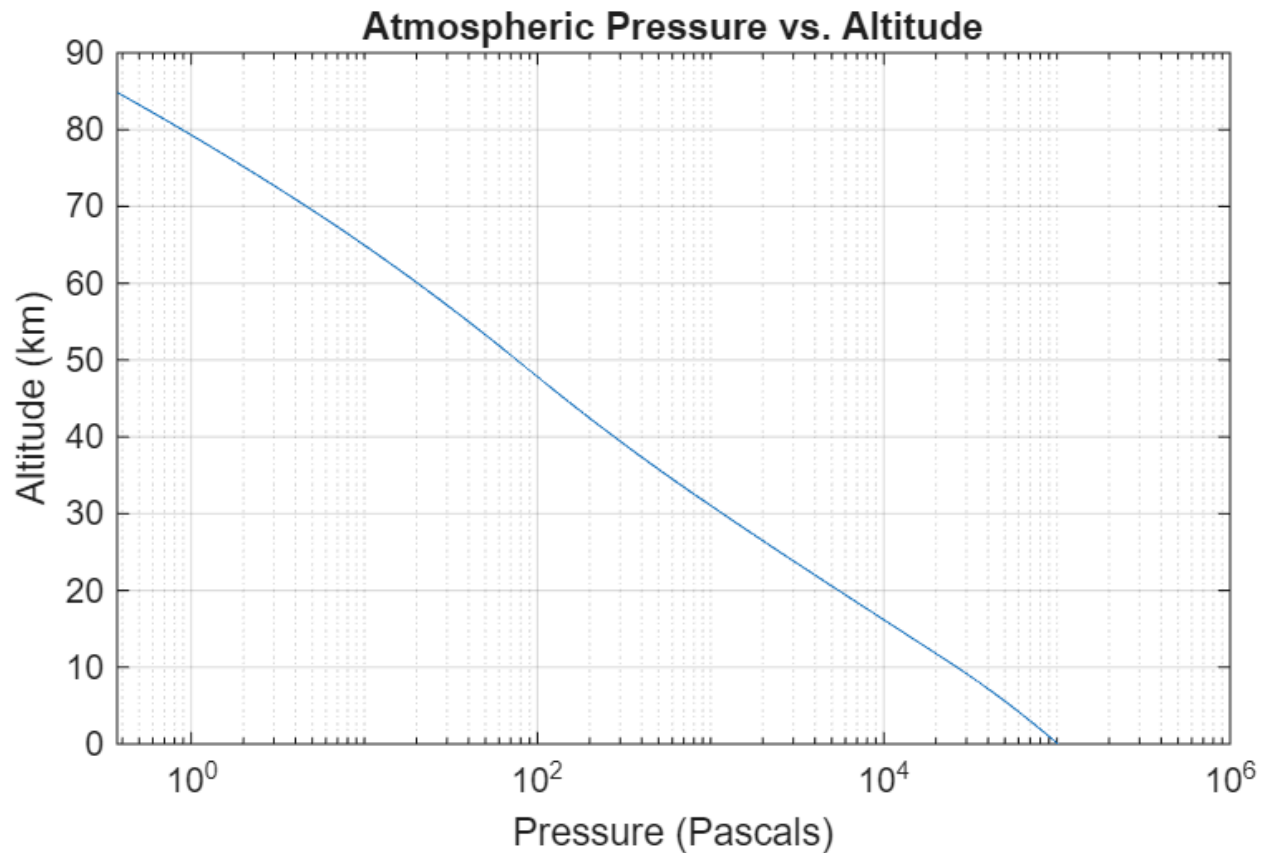
$$P(H) = P_0 \cdot e^{\left(\frac{g_0 \cdot M}{R \cdot L_0} \right)}$$

Equation for finding the pressure assuming no temperature gradient

Where P_0 is the reference pressure at sea level, T_0 and H_0 are the reference temperature and altitude at the start of the lapse region, and L is the lapse rate. The

exponent term g_0 is the gravitational acceleration, M is the mean molar mass of Earth's air, and R is the universal gas constant.

Using these equations, we were able to find the following pressure profile. We used a semi-log x plot to better showcase this relation.

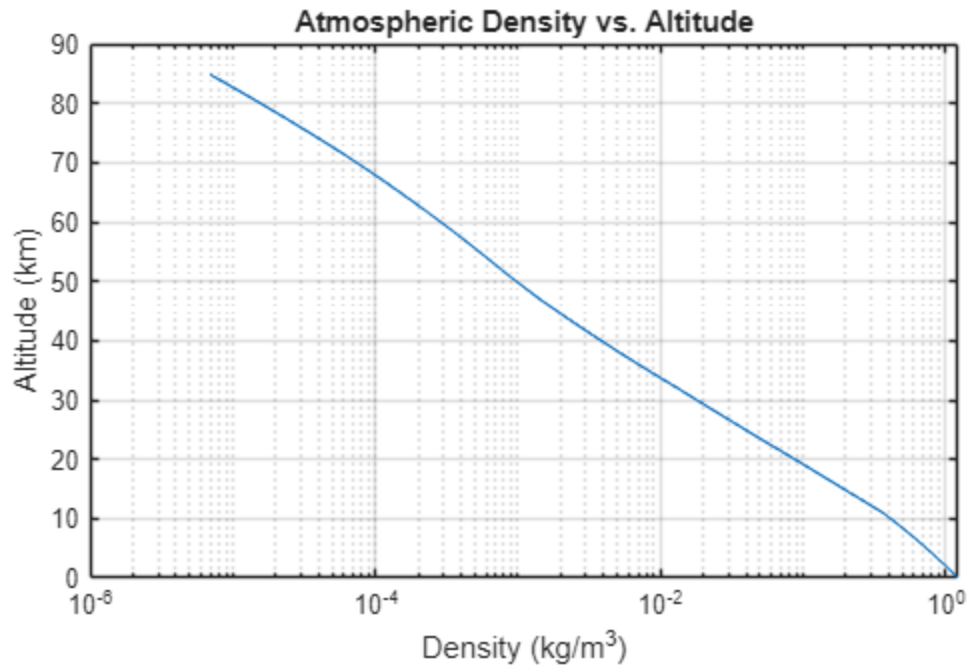


Once pressure and temperature are known, density is calculated from the ideal gas law:

$$\rho = \frac{P}{R_{specific}T}$$

In this equation, ρ is air density, P is pressure, $R_{specific}$ is the specific gas constant for air, and T is temperature.

This atmospheric model provides the density data needed for the trajectory and heating calculations. The density plot below also uses a semi-log x plot to showcase the relation more clearly.



3.2 Trajectory Model

After building the atmospheric model, we simulated a simplified spacecraft descent. The spacecraft was treated as a point mass with constant mass, drag coefficient, cross-sectional area, and flight-path angle. The initial conditions were set to represent a vehicle beginning re-entry at high altitude and hypersonic speed.

The key vehicle parameters used in the model were:

Mass: 9298.64 kg

Drag coefficient: 1.2

Cross-sectional area: 19.87 m^2

Flight-path angle: 8 degrees

Initial velocity: 7800 m/s

Initial altitude: 84852 m

The key heat shield parameters were:

Emissivity of the heatshield: 0.9

Ablation temperature: 1973.15 K

Effective heat of ablation: $1.5 \cdot 10^7$ J/kg

Curvature of heatshield: 20 degrees

We collected as many values as we could from the Atrimis II mission and filled in the other parameters from other similar spacecraft missions. One key note is that we were not able to find the curvature of the heat shield, so we created a reasonable approximation from available visuals.

For the Sutton-Graves Correlation, we have found the changing density with altitude, we know the effective nose radius, and the planetary atmospheric constant, so all that we are missing is velocity. To find the velocity, we use the starting entry velocity and the deceleration the aircraft experiences as it goes through the atmosphere. This specific equation is part of a set of planar reentry equations of an unpowered spacecraft (basically just a meteor).

$$a = g \sin(\gamma) - \frac{\rho V^2}{2\beta}$$

In this equation, a is acceleration, g is gravitational acceleration, γ is the flight-path angle, ρ is local atmospheric density, V is spacecraft velocity, and β is the ballistic coefficient. And the altitude decreases by multiplying the negative of velocity by the sine of the flight angle.

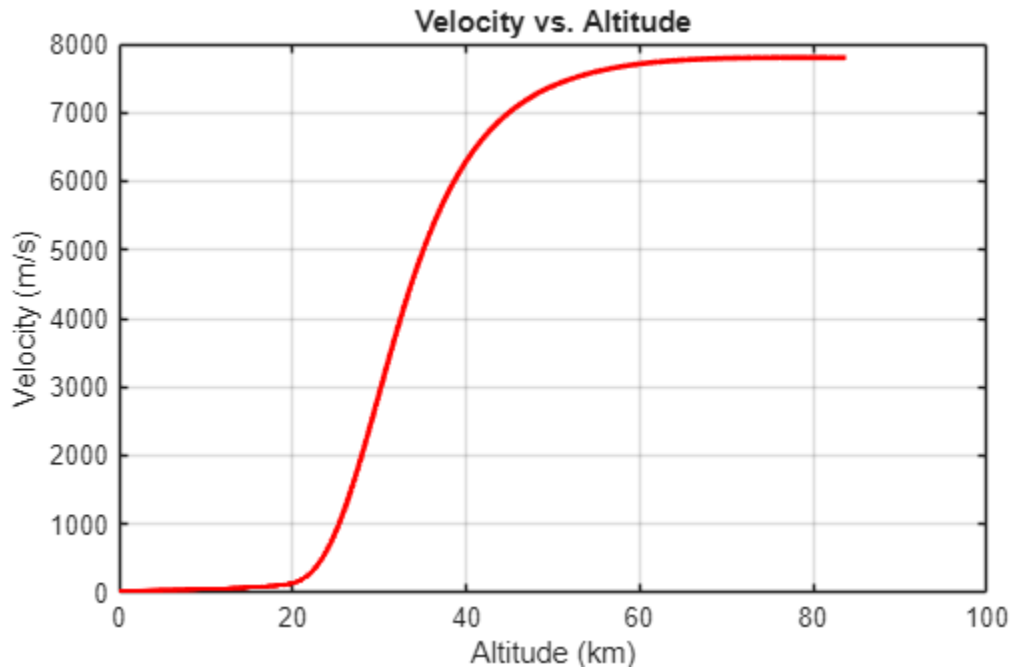
The ballistic coefficient is given by the following equation:

$$\beta = \frac{m}{C_d A}$$

In this equation, β is the ballistic coefficient, m is the spacecraft mass, C_d is the drag coefficient, and A is the cross-sectional area.

The velocity and altitude were then updated step by step over time. This produced a trajectory showing how the spacecraft slows down as it descends into denser

layers of the atmosphere. The code outputs a plot of velocity vs. altitude, which illustrates the deceleration behavior during re-entry.



3.3 Aerodynamic Heating Model

The heating model estimates the convective stagnation-point heat flux using the Sutton-Graves relation:

$$q''_{\text{stagnation}} = k_{sg} \sqrt{\frac{\rho}{R_n}} V^3$$

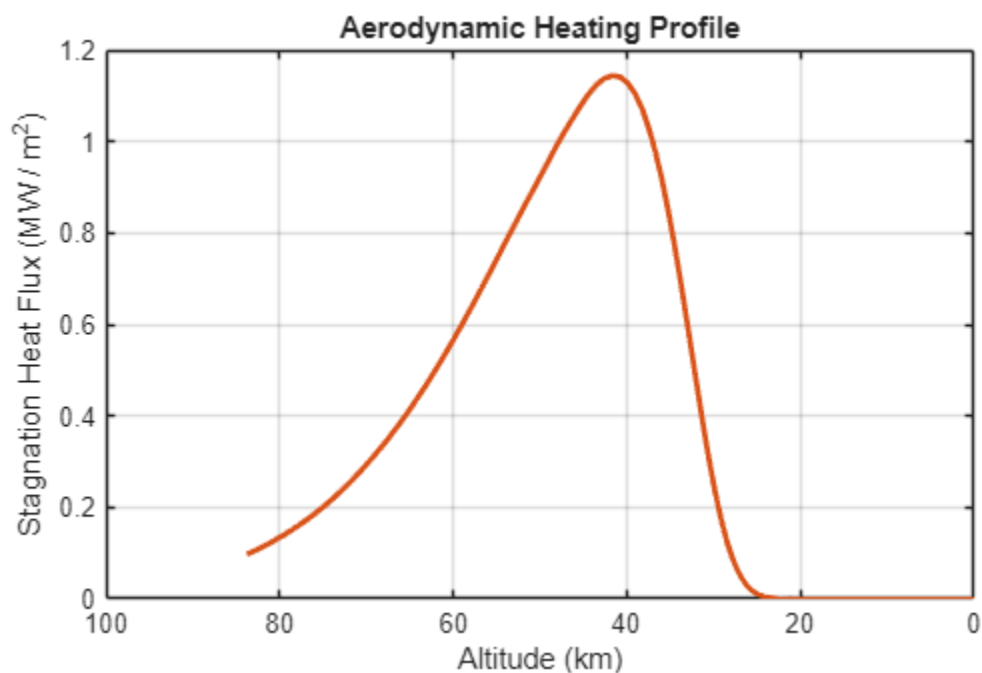
In this equation, $q''_{\text{stagnation}}$ is the aerodynamic stagnation-point heat flux, k_{sg} is the Sutton-Graves constant for Earth, ρ is atmospheric density, R_n is the nose radius of the spacecraft, and V is spacecraft velocity.

For this model, the assumed nose radius was 2.51 m, and the Sutton-Graves constant was:

$$k_{sg} = 1.7415 \times 10^{-4}$$

This equation shows an important physical result: aerodynamic heating depends extremely strongly on velocity because the velocity term is cubed. Even if atmospheric density is still low at high altitude, the high re-entry speed can produce substantial heating. As the spacecraft descends, density increases while velocity decreases, creating a heating profile that changes significantly throughout the descent.

The code produces a plot of stagnation heat flux vs. altitude. This plot shows how the aerodynamic heating changes over the trajectory. It also produces a surface energy balance plot that currently includes the incoming aerodynamic heat term.

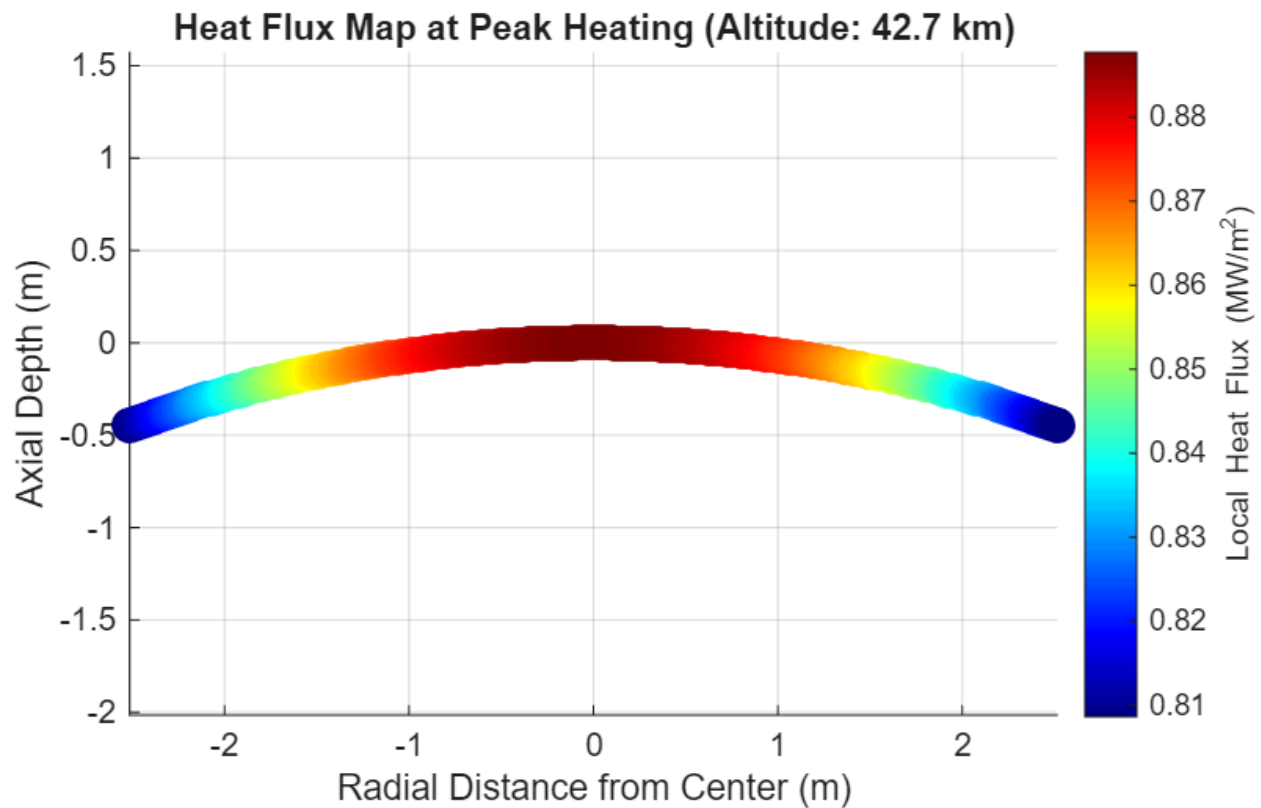


3.4 Heat Flux Distribution

While the correlation provides the absolute maximum heating experienced by the spacecraft, that only applies at a single point on near the center of the spacecraft's nose. It is important to find out how the heat is distributed along the nose of the spacecraft, as that will ultimately dictate the amount of ablation that occurs and the needed thickness of the heat shield. To accurately model the heat across the entire curved surface of a blunt body, we utilize Lee's Heat Transfer Distribution:

$$q''(\theta) = q''_{stagnation} \cdot \cos^{3/2}(\theta)$$

Where $q''(\theta)$ is the local heat flux at a specific point on the heat shield and $q''_{\text{stagnation}}$ is the maximum heating from the Sutton Graves equation, and θ is the angle measured from the center vertical axis of the spacecraft.



This equation highlights a critical fluid dynamics concept: the hypersonic airflow slides along the curved heatshield and expands rapidly, reducing the heat generated at the more vertical edges.

3.5 Surface Energy Balance and Mass Ablation

The final stage of the model determines how much of the heat shield physically gets burned away. The shield has two primary heat dissipation methods, radiation and ablation. Before the heat shield physically melts, it glows red and acts as a nearly perfect blackbody radiator, radiating away nearly all of the heat. We modeled this out-going radiative heat as:

$$q_{rad} = \epsilon \cdot \sigma \cdot (T_{ablation})^4$$

In this equation ϵ is the emissivity (set at 0.9), σ is the Boltzmann constant, and T_{ablation} is the failure temperature of the material (1973.15 K).

One crucial assumption to note is that we are assuming that heat is only being radiated away, and the rest of the heat that is larger than the q_{rad} limit is converted to ablative heat that eats away at the heat shield. This was done since this is only a 2-week project, and modeling the heat that gets transferred to the heat shields, and then gets transferred to the Inconel bolts and bushings to find how much the cabin heats up is outside of the scope for this project.

Only the heat that overpowers the radiative limit actually causes physical material damage:

$$q_{net} = q''(\theta) - q_{rad}$$

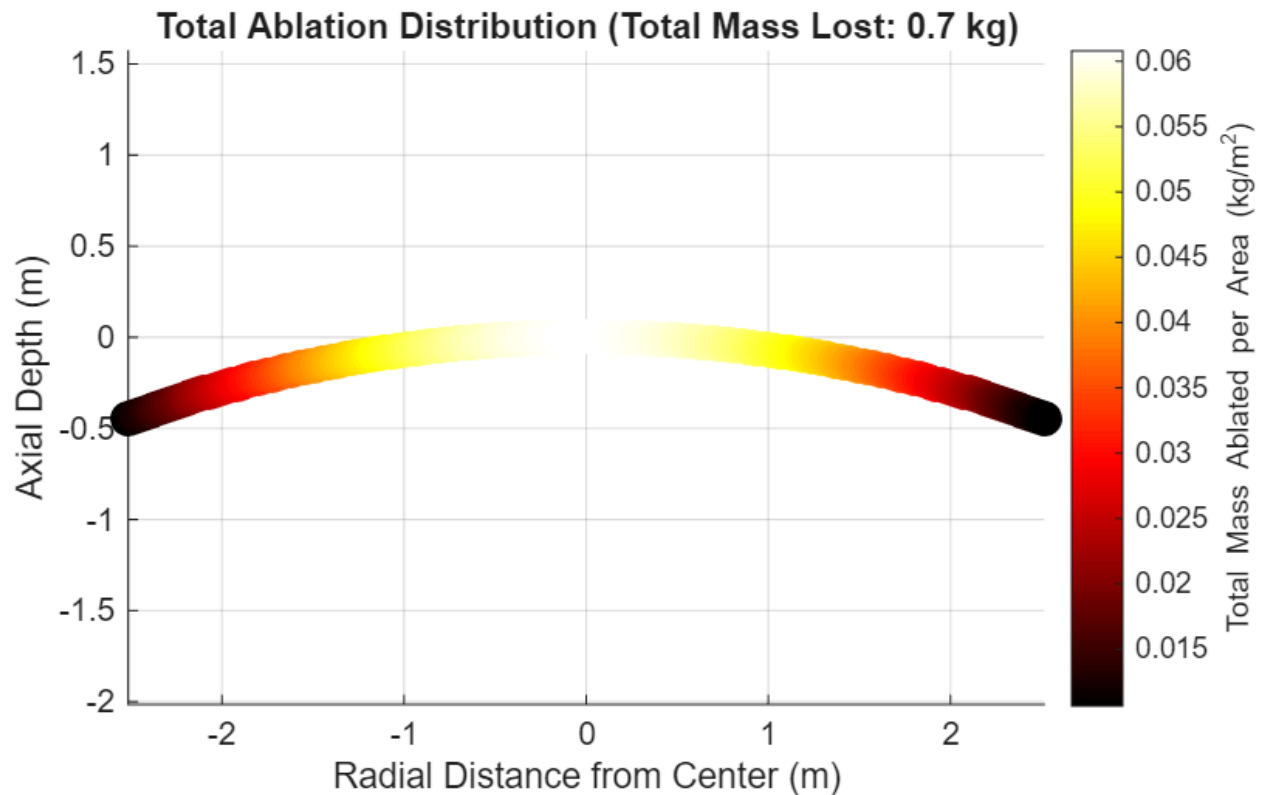
Where $q''(\theta)$ is the heat flux distribution, q_{rad} is the maximum radiative heat, and q_{net} is the ablative heat flux. Note that ablation only occurs when q_{net} is positive. If q_{net} is negative, then the heat shield has successfully radiated all incoming heat away.

When ablation occurs, the mass loss rate for each very small ring approximation is calculated using the Effective Heat of Ablation.

$$m_{ring} = \frac{q_{net} \cdot A_{ring}}{H_{eff}}$$

Where q_{net} is the previously found positive ablative heat flux, A_{ring} is the area of each small ring, and H_{eff} is the effective heat of ablation (which is the heat required to vaporize 1 kg of material) ($1.5 \cdot 10^7 \text{ J/kg}$).

During our simulation loop, our code calculates the net heat and resulting mass loss for all 100 rings at every time step and sums these values together to output the total ablative mass loss. Our code also produces a heat map showcasing which rings yielded the most mass loss rate, shown below.



4. Conclusion

Spacecraft re-entry is a complex thermal-fluid problem where hypersonic motion, atmospheric density, shock layer heating, radiation, and material ablation all interact. In this project, we first learned the main physics behind re-entry heating and then built a simplified model to connect those ideas to heat shield performance. Our model showed that heating is not constant throughout descent. Instead, the maximum heating occurs when the spacecraft is still moving very fast but has entered denser parts of the atmosphere. This matches the physical idea that stagnation-point heat flux depends strongly on both velocity and atmospheric density.

Using the Sutton-Graves correlation, we estimated the stagnation-point heating during descent. Then, using Lee's heat flux distribution, we showed how the heat is spread across the curved heat shield instead of being equal everywhere. The center region near the stagnation point experienced the highest heating, while the outer edges experienced less heating because the flow expands along the curved surface. Finally, we used a simplified surface energy balance to estimate how much material would be lost after radiation was accounted for. In our model, the total predicted ablation was about 0.7 kg.

Overall, this project helped us see that a heat shield is not simply a barrier that blocks heat. It is an energy-management system that protects the spacecraft by spreading heat, radiating energy away, and sacrificing material through ablation when necessary. Although our model makes many simplifications, it captures the main relationships between altitude, velocity, density, heat flux, and ablation. A more complete model would include changing material properties, conduction into the spacecraft, surface recession, pyrolysis gases, and a more realistic trajectory, but our simplified approach still gives a useful first-order understanding of why heat shield design is so important during re-entry.

5. List of Assumptions and Limitations

- Sutton-Graves Correlation
 - **Hypersonic Flow:** Flow is greater than Mach 5 (1715 m/s)
 - **Continuum Flow:** Air is smooth and continuous (no molecular physics)
 - **Thermal and Chemical Equilibrium:** All energy modes and chemical reactions have reached a steady state
 - **Blunt Body Geometry:** Most accurate for blunt bodies, where the leading edge radius is large compared to the boundary layer thickness.
- Barometric Formula
 - Valid from sea level to an altitude of **86 km** above sea level.
- Planar Re-entry equations
 - **Entry is purely ballistic** (zero produced lift)
 - The effect of **gravity can be ignored** since it is small compared to drag.
 - Flight angle and ballistic coefficients are constant
- General Core Model Assumptions
 - Assumed not a lot of variation between different spacecrafts, as no one spacecraft had ALL the information we needed.
 - **Constant Gravity**
 - **Flat not rotating earth** (not accounting for Coriolis effects)
 - **Quasi-Steady State Ablation:** Heat shield starts radiating heat assuming it is constantly at its ablation temperature, no conduction term, assuming all heat is either radiative or ablative (worst case scenario).
 - **Constant material properties:** Emmisivity and effective heat of ablation are assumed to be constant
 - **1D Independent Heat Transfer:** Assumes heat only travels straight into material, ignoring lateral heat transfer.

6. Sources

[1] Wikipedia, "Planar reentry equations," Wikimedia Foundation. [Online].

https://en.wikipedia.org/wiki/Planar_reentry_equations

[2] NASA, "Spacecraft Components," NASA. [Online].

<https://www.nasa.gov/reference/spacecraft-components/>

[3] Emisshield, "Emisshield Products," Emisshield Inc. [Online].

<https://emissshield.com/emissshield-products/>

[4] Wikipedia, "Barometric formula," Wikimedia Foundation. [Online].

https://en.wikipedia.org/wiki/Barometric_formula

[5] M. A. Covington, "Performance of a Light-Weight Ablative Thermal Protection Material For the Stardust Mission Sample Return Capsule," NASA Technical Reports Server, 2007.

<https://ntrs.nasa.gov/api/citations/20070014634/downloads/20070014634.pdf>

[6] J. A. Dec and R. D. Braun, "An Approximate Ablative Thermal Protection System Sizing Tool for Entry System Design," 44th AIAA Aerospace Sciences Meeting, 2006.

<https://ntrs.nasa.gov/api/citations/20060004824/downloads/20060004824.pdf>

[7] Federal Aviation Administration, "Returning from Space," FAA. [Online].

https://www.faa.gov/sites/faa.gov/files/about/office_org/headquarters_offices/avs/III.4.1.7_Returning_from_Space.pdf

[8] K. Bibb, E. Walker, G. Brauckmann, and P. Robinson, "Development of the Orion Crew Module Static Aerodynamic Database," 29th AIAA Applied Aerodynamics Conference, 2011. [Online].

<https://scispace.com/pdf/development-of-the-orion-crew-module-static-aerodynamic-2hy6t1ls9c.pdf>

[9] R. Kumar, "Numerical Investigation of Gas-Surface Interactions due to Ablation of High-Speed Vehicles," Journal of Spacecraft and Rockets, vol. 53, no. 3, pp. 538–548, 2016.

<https://arc.aiaa.org/doi/epdf/10.2514/1.A33433>

[10] B. M. Babu, M. K. P., R. M. Robinson, V. K. K., and D. Deepak, "Design of Thermal Protection System for Low Ballistic Coefficient Ballute for Ballistic Earth Re-entry," International Research Journal of Engineering and Technology (IRJET), vol. 7, no. 9, Sep. 2020. [Online].

<https://issuu.com/irjet/docs/irjet-v7i936/s/11613760>